

Solution of Tutorial-4

Engineering Mechanics: ME-101 (Semester-II, 2024)

Time: 7:55 – 8:50 AM

Q1. Determine the shear force and the moment at point C of the beam shown in Figure 1.

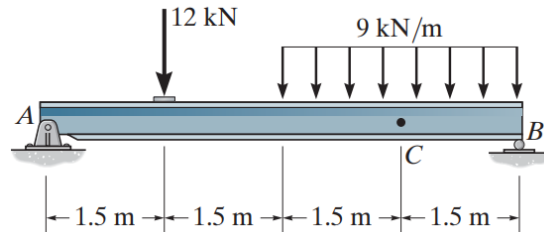
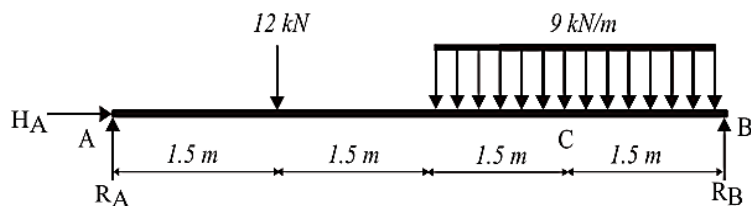


Figure 1

Solution of Q1:

Draw the FBD for the beam



(1 mark)

Step 1: Finding out the support reactions

$\sum F_x = 0$ leads to $H_A = 0$ (Since there is no horizontal loading) (1 mark)

$\sum F_y = 0$ (Considering upward forces positive)

$$R_A + R_B = 12 + (9 \times 3) = 39 \text{ kN} \quad \text{-----(1)}$$

$\sum M_A = 0$ (Considering counterclockwise moments positive)

$$R_B \times 6 - (9 \times 3 \times 4.5) - (12 \times 1.5) = 0 \quad \text{-----(2)}$$

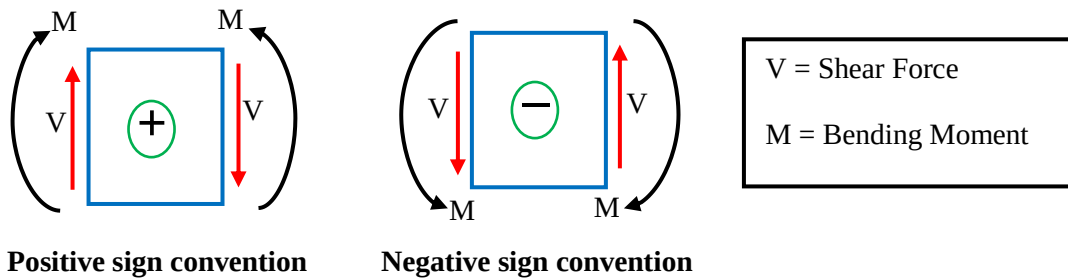
By solving equations (1) and (2) we obtain,

$$R_B = 23.25 \text{ kN} \quad (1 \text{ mark})$$

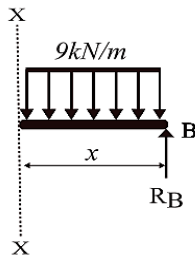
$$R_A = 15.75 \text{ kN} \quad (1 \text{ mark})$$

Step 2: Finding out the Shear force and Bending moment at point C

First, let us have a look at the sign convention of SF and BM followed throughout



Let's consider a section X – X, within BC at a distance x from point B,



Force and moment equilibrium equations yields,

$$V_x + R_B - (9 \times x) = 0 \rightarrow V_x = -R_B + 9x \quad (1 \text{ mark})$$

$$M_x - R_B \times x - (9 \times x) \frac{x}{2} = 0 \rightarrow M_x = R_B x + \frac{9x^2}{2} \quad (1 \text{ mark})$$

Shear force and Bending moment at $x = 0$, that is at point B

$$V_B = -23.25 + (9 \times 0) = -23.25 \text{ kN} \quad (1 \text{ mark})$$

$$M_B = 23.25 \times 0 - \frac{9 \times 0^2}{2} = 0 \quad (1 \text{ mark})$$

Shear force and Bending moment at $x = 1.5 \text{ m}$, that is at point C

$$V_C = -23.25 + (9 \times 1.5) = -9.75 \text{ kN} \quad (1 \text{ mark})$$

$$M_C = 23.25 \times 1.5 - \frac{9 \times 1.5^2}{2} = 24.75 \text{ kNm} \quad (1 \text{ mark})$$

Q2. Figure 2 shows the beam loaded with both the distributed and point loads.

- Calculate of support reaction at point A and B. [1 mark]
- Write the shear force and bending moment equations for any point in AC portion of the beam in terms of x , where x is distance from point A [1 mark]
- Write the shear force and bending moment equations for any point in CD portion of the beam in terms of x , where x is distance from point A [1 mark]
- Write the shear force and bending moment equations for any point in DE portion of the beam in terms of x , where x is distance from point A [1 mark]
- Write the shear force and bending moment equations for any point in EB portion of the beam in terms of x , where x is distance from point A [1 mark]
- Calculate shear force and bending moment at $x = 6 \text{ m}$. [1 mark]
- Calculate maximum bending moment and location of maximum bending moment from point A. [2 marks]
- Plot shear force and bending moment diagram for entire beam. [2 marks]

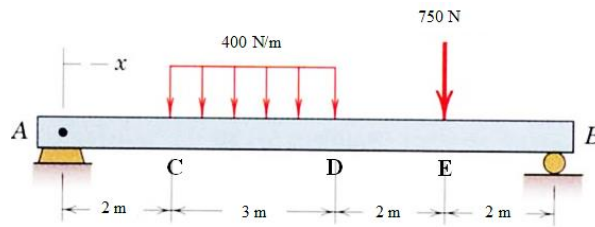


Figure 2

Solution of Q2:

Let us consider upward and rightward direction force and counter clock wise moment as positive.

Step 1: Calculation of support reactions

[1 mark]

Free body diagram of entire beam

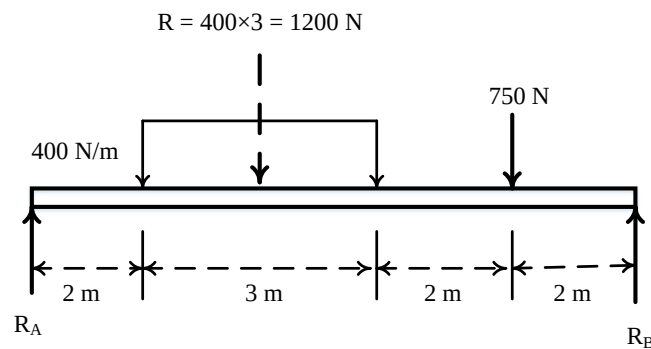


Figure 2.1

Calculation of support reaction force at point A and B

Taking moment about point A

$$\begin{aligned}\sum M_A &= 0 \\ \therefore R_B \times 9 - 750 \times 7 - 1200 \times 3.5 &= 0 \\ R_B &= 1050 \text{ N}\end{aligned}$$

By force balance equation in y-direction:

$$\begin{aligned}\sum F &= 0 \\ \therefore R_A - 1200 - 750 + 1050 &= 0 \\ R_A &= 900 \text{ N}\end{aligned}$$

Step 2: Calculation of shear force and bending moment between $0 < x < 5 \text{ m}$ [2 marks]

Section 1: $0 < x < 2 \text{ m}$:

From left side upward direction force and anti-clock wise moment is positive

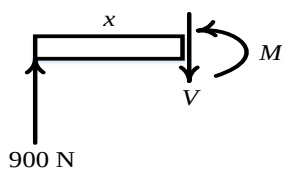


Figure 2.2

By force balance equation in y-direction the shear force V can be calculated as follows

$$\begin{aligned}\sum F &= 0 \\ V - 900 &= 0 \\ V &= 900 \text{ N}\end{aligned}$$

Taking moment about the point of section, the bending moment M can be calculated as follows:

$$\begin{aligned}\sum M &= 0 \\ M - 900x &= 0\end{aligned}$$

$$M = 900x$$

Section 2: $2 < x < 5$ m: From left side upward direction force and anti-clock wise moment is positive

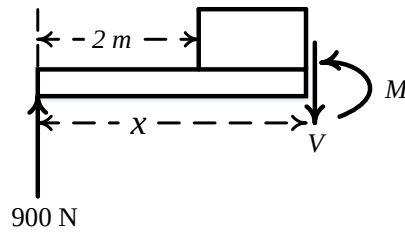


Figure 2.3

By force balance equation in y-direction the shear force V can be calculated as follows:

$$\begin{aligned}\sum F &= 0 \\ 900 - 400(x - 2) - V &= 0 \\ V &= 1700 - 400x\end{aligned}$$

Taking moment about the point of section, the bending moment M can be calculated as follows:

$$\begin{aligned}\sum M &= 0 \\ M + 400(x - 2)\frac{(x - 2)}{2} - 900x &= 0 \\ M &= -200x^2 + 1700x - 800\end{aligned}$$

Step 3: Calculation of shear force and bending moment between $5 < x < 7$ m [2 marks]

Section 3: $5 < x < 7$ m:

From right side upward direction force and clock wise moment is positive

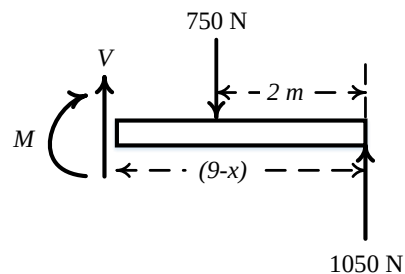


Figure 2.4

By force balance equation in y-direction the shear force V can be calculated as follows:

$$\begin{aligned}\sum F &= 0 \\ 1050 - 750 + V &= 0 \\ V &= -300 \text{ N}\end{aligned}$$

Taking moment about the point of section, the bending moment M can be calculated as follows:

$$\begin{aligned}\sum M &= 0 \\ -M - 750(7 - x) + 1050(9 - x) &= 0 \\ M &= 4200 - 300x\end{aligned}$$

Section 4: $7 < x < 9$ m:

From right side upward direction force and clock wise moment is positive

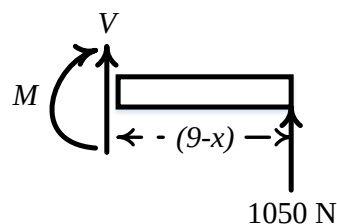


Figure 2.5

By force balance equation in y-direction the shear force V can be calculated as follows:

$$\begin{aligned}\sum F &= 0 \\ V &= -1050 \text{ N} \\ \sum M &= 0 \\ M &= 9450 - 1050x\end{aligned}$$

Step 4: Calculation of shear force and bending moment at $x = 6 \text{ m}$

[1 mark]

For shear force and bending moment at $x = 6 \text{ m}$:

From section 3

$$\begin{aligned}V &= -300 \text{ N} \\ M &= 4200 - 300(6) \\ M &= 2400 \text{ N.m}\end{aligned}$$

Step 5: Calculation of maximum bending moment and location of bending moment from support A

[2 marks]

We know that the maximum bending moment occurs at a point on a beam where the shear force changes its sign. The bending moment is also maximum when the shear force is zero.

From shear force diagram we have seen that shear force changes its sign in section 2, hence

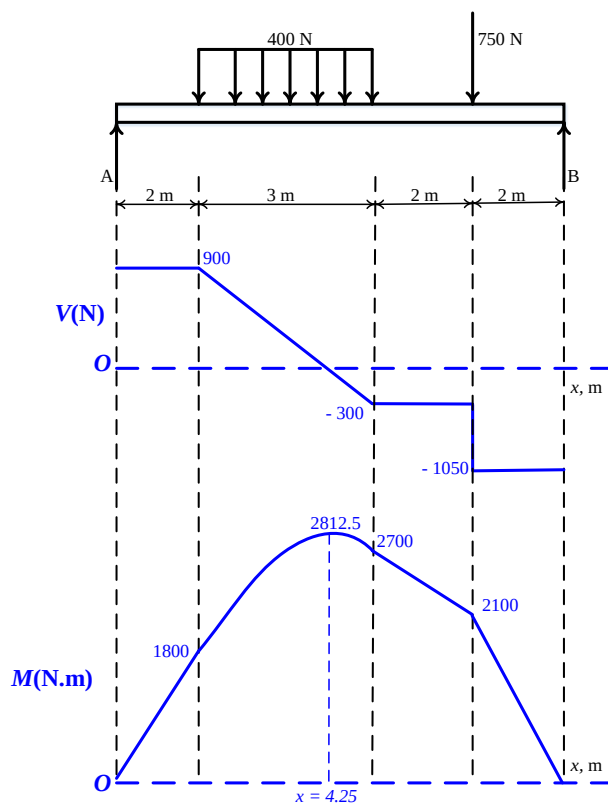
$$\begin{aligned}\frac{dM}{dx} &= 0 \\ \frac{d}{dx}(-200x^2 + 1700x - 800) &= 0 \\ \therefore -400x + 1700 &= 0 \\ x &= 4.25 \text{ m}\end{aligned}$$

So, we can say that maximum bending moment in the beam is present at a distance of $x = 4.25 \text{ m}$ from point A.

$$M_{max} = -200(4.25)^2 + 1700(4.25) - 800 = 2812.5 \text{ N.m}$$

Step 6: Shear force and Bending moment diagram for entire beam

[2 marks]



Q3. For the beam and loading shown in Figure 2, draw the shear force and bending moment diagrams. Determine the maximum values of shear force and bending moment.

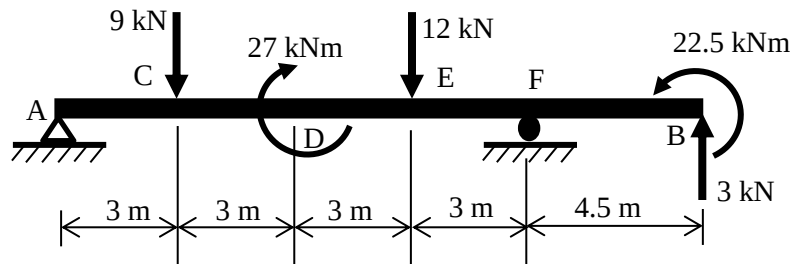
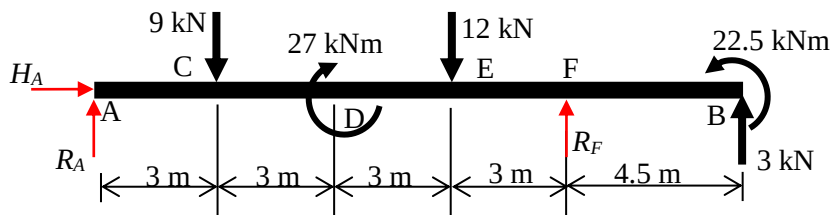


Figure 3

Solution of Q3:

Draw the FBD for the beam



The given beam is statically determinate since the no. of support reactions is 3 and the available equations for static equilibrium are also 3.

Step 1: Finding out the support reactions

Since, there is no horizontal loading, $\sum F_x = 0$ leads to $H_A = 0$

Now, $\sum F_y = 0$, (considering upward forces positive)

$$\Rightarrow R_A + R_F + 3 - 9 - 12 = 0 \Rightarrow R_A + R_F = 18 \quad \text{----- (1)}$$

$\sum M_A = 0$, (considering counter clockwise moments positive)

$$\Rightarrow R_F \times 12 + 3 \times 16.5 + 22.5 - 12 \times 9 - 27 - 9 \times 3 = 0 \Rightarrow R_F = 7.5 \text{ kN}$$

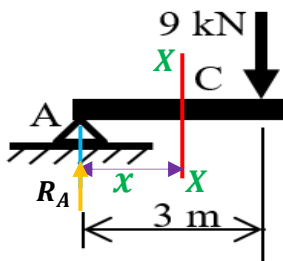
From equation (1), we obtain $R_A = 10.5 \text{ kN}$

(1 mark)

Step 2: Finding out the SF and BM

The beam AB has 5 segments AC, CD, DE, EF, and FB. We will find out the shear forces and bending moments by taking a section X – X at a distance x from A with varying limits in all these segments.

Segment AC



Section X – X at a distance x ($0 \leq x \leq 3$) from A

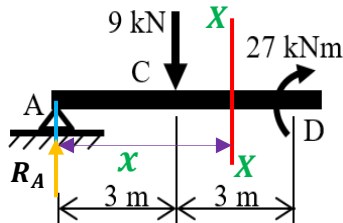
SF at section $X - X = R_A$

BM at section $X - X = R_A \times x$

Now, $SF_{x=0} = 10.5 \text{ kN}$ & $SF_{x=3} = 10.5 \text{ kN}$

$BM_{x=0} = 0 \text{ kNm}$ & $BM_{x=3} = 10.5 \times 3 = 31.5 \text{ kNm}$ (1.5 marks)

Segment CD



Section $X - X$ at a distance x ($3 \leq x \leq 6$) from A

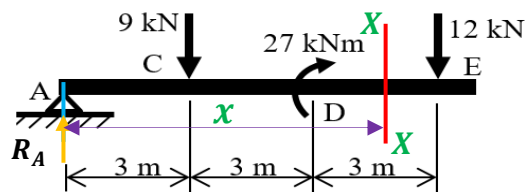
SF at section $X - X$, $V_x - R_A + 9 = 0 \rightarrow V_x = R_A - 9$

BM at section $X - X$, $M_x - R_A \times x + 9(x - 3) = 0 \rightarrow M_x = R_A \times x - 9(x - 3)$

Now, $SF_{x=3} = 1.5 \text{ kN}$ & $SF_{x=6} = 1.5 \text{ kN}$

$BM_{x=3} = 31.5 \text{ kNm}$ & $BM_{x=6} = 36 \text{ kNm}$ (1.5 marks)

Segment DE



Section $X - X$ at a distance x ($6 \leq x \leq 9$) from A

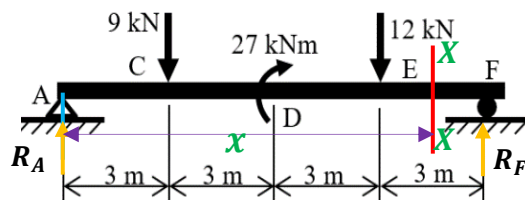
SF at section $X - X$, $V_x - R_A + 9 = 0 \rightarrow V_x = R_A - 9$

BM at section $X - X$, $M_x - R_A \times x + 9(x - 3) - 27 = 0 \rightarrow M_x = R_A \times x - 9(x - 3) + 27$

Now, $SF_{x=6} = 1.5 \text{ kN}$ & $SF_{x=9} = 1.5 \text{ kN}$

$BM_{x=6} = 63 \text{ kNm}$ & $BM_{x=9} = 67.5 \text{ kNm}$ (1.5 marks)

Segment EF



Section $X - X$ at a distance x ($9 \leq x \leq 12$) from A

SF at section $X - X$ $V_x - R_A + 9 + 12 = 0 \rightarrow V_x = R_A - 21$

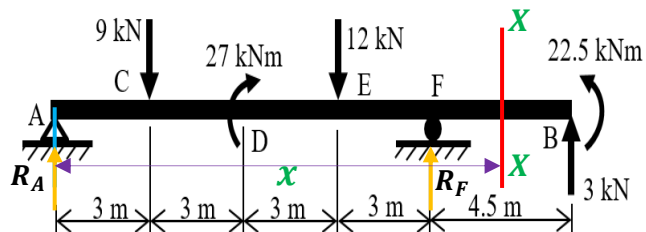
$$BM \text{ at section } X-X \quad M_x - R_A \times x + 9(x-3) - 27 + 12(x-9) = 0$$

$$\text{Now, } SF_{x=9} = -10.5 \text{ kN} \text{ \& } SF_{x=12} = -10.5 \text{ kN}$$

$$BM_{x=9} = 67.5 \text{ kNm} \text{ \& } BM_{x=12} = 36 \text{ kNm}$$

(1.5 marks)

Segment FB



Section X – X at a distance x ($12 \leq x \leq 16.5$) from A

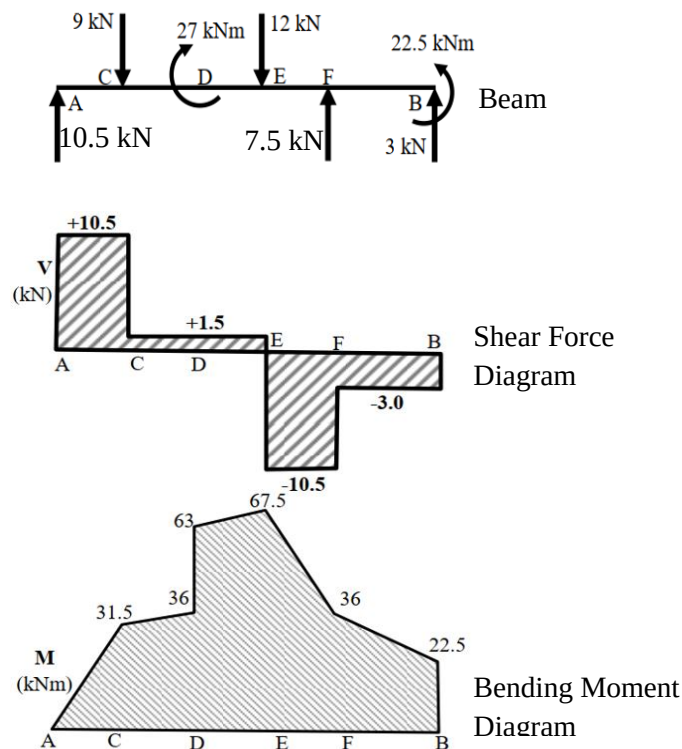
$$SF \text{ at section } X-X, V_x - R_A + 9 + 12 - R_F = 0 \rightarrow V_x = R_A - 21 + R_F$$

$$BM \text{ at section } X-X, M_x - R_A \times x + 9(x-3) - 27 + 12(x-9) - R_F \times (x-12) = 0 \rightarrow M_x = R_A \times x - 9(x-3) + 27 - 12(x-9) + R_F \times (x-12)$$

$$\text{Now, } SF_{x=12} = -3 \text{ kN} \text{ \& } SF_{x=16.5} = -3 \text{ kN}$$

$$BM_{x=12} = 36 \text{ kNm} \text{ \& } BM_{x=16.5} = 22.5 \text{ kNm}$$

(1.5 marks)



(1 mark)

From the above diagram it is clear that the max magnitude of the shear force V_{max} is 10.5 kN in the interval AC and EF, whereas the max magnitude of bending moment M_{max} is 67.5kN-m at E.

(0.5 mark)

Q4. A simply supported beam with two equal overhangs is loaded by uniformly distributed load w on entire length (see Figure 3). If support position is adjustable then draw the shear force and bending moment diagrams for following three cases: **a)** $l > 2a$; **b)** $l = 2a$; and **c)** $l < 2a$.

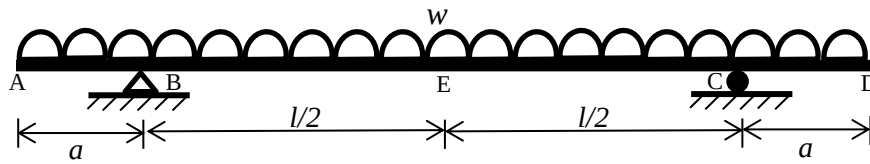
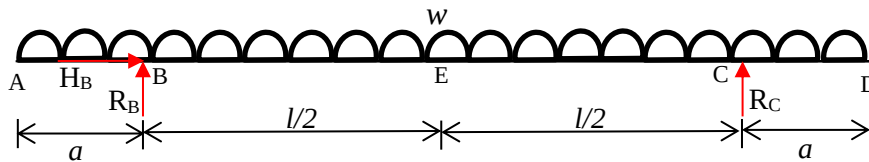


Figure 4

Solution of Q4:

Draw the FBD for the beam



Step 1: Finding out the support reactions

Since, there is no horizontal loading, $\sum F_x = 0$ leads to $H_B = 0$

$$\sum F_y = 0 \quad (\text{Considering upward forces positive})$$

$$R_C + R_B = w \times (l + 2a) \quad \text{-----}(1)$$

$$\sum M_B = 0 \quad (\text{Considering counterclockwise moments positive})$$

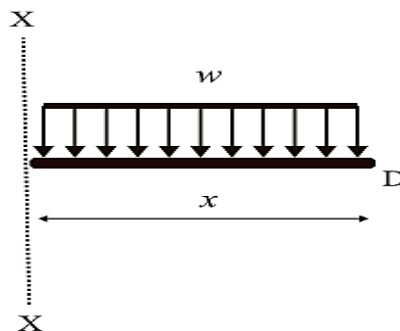
$$-w \times (l + 2a) \times \frac{l}{2} + R_C \times l = 0 \quad \text{-----}(2)$$

By solving equations 1 and 2

$$R_B = R_C = w \left(\frac{l}{2} + a \right)$$

(1 mark)

Step 2: Finding out Shear forces and Bending moments



Let's cut a section X – X in the segment CD ($0 \leq x \leq a$) and consider x from point D

Shear force and bending moment in this particular section are found to be,

$$V_x - w \times x = 0$$

$$M_x + w \times x \times \left(\frac{x}{2} \right) = 0$$

Shear force and Bending moment at $x = 0$, that is at point D

$$V_D = w \times (0) = 0$$

(1 mark)

$$M_D = -\frac{w \times 0^2}{2} = 0$$

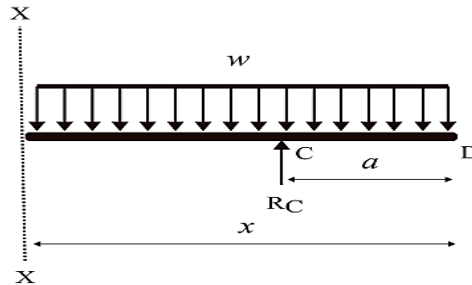
Shear force and Bending moment at $x = a$, that is at point C (Just before C)

$$V_C = w \times (a) = wa$$

(1 mark)

$$M_C = -\frac{wa^2}{2} \text{ (Just before C)}$$

Let's cut a section X – X in the segment CB ($a \leq x \leq (l + a)$) and consider x from point D



Shear force and bending moment equations in this particular section

$$V_x - wx + R_C = 0$$

$$\hookrightarrow V_x = wx - \left(\frac{wl}{2} + wa\right)$$

(1 mark)

$$M_x + wx\left(\frac{x}{2}\right) - R_C \times (x - a) = 0$$

$$\hookrightarrow M_x = -\left(\frac{wx^2}{2}\right) + \left(\frac{wl}{2} + wa\right) \times (x - a)$$

Shear force and Bending moment at $x = a$, that is at point C

$$V_C = wa - \left(\frac{wl}{2} + wa\right) = -\frac{wl}{2} \text{ (Just after C from right)}$$

(1 mark)

$$M_C = -\frac{wa^2}{2} \text{ (Just after C from right)}$$

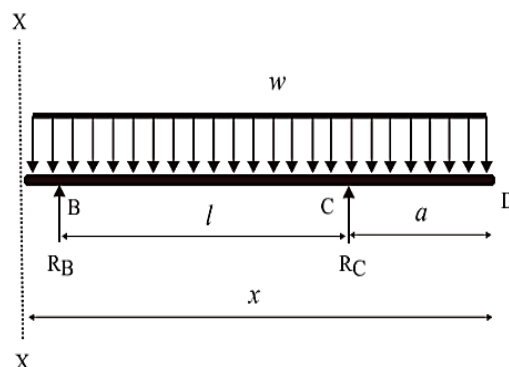
Shear force and Bending moment at $x = (l + a)$, that is at point B

$$V_B = w(l + a) - \left(\frac{wl}{2} + wa\right) = \frac{wl}{2} \text{ (Just before B from right)}$$

(1 mark)

$$M_B = -\left(\frac{w(l+a)^2}{2}\right) + \left(\frac{wl}{2} + wa\right) \times ((l + a) - a) = -\frac{wa^2}{2} \text{ (Just before B from right)}$$

Let's cut the section X – X in the segment BA ($(l + a) \leq x \leq (l + 2a)$) and consider x from point D



Shear force and bending moment equations in this particular section

$$V_x - wx + \left(\frac{wl}{2} + wa\right) + \left(\frac{wl}{2} + wa\right) = 0 \Rightarrow V_x = \frac{2wx - 2wl - 4wa}{2}$$

$$M_x + \left(\frac{wx^2}{2}\right) - \left(\frac{wl}{2} + wa\right) \times (x - a) - \left(\frac{wl}{2} + wa\right) \times (x - (l + a)) = 0$$

$$\hookrightarrow M_x = -\left(\frac{wx^2}{2}\right) + \left(\frac{wl}{2} + wa\right) \times (x - a) + \left(\frac{wl}{2} + wa\right) \times (x - (l + a))$$

Shear force and Bending moment at $x = (l + a)$, that is at point B

$$V_B = \frac{2w(l+a) - 2wl - 4wa}{2} = -wa \text{ (Just after B)}$$

(1 mark)

$$M_B = -\frac{wa^2}{2} \text{ (Just after B)}$$

Shear force and Bending moment at $x = (l + 2a)$, that is at point A

$$V_A = 0$$

$$M_A = 0$$

Bending moment at $x = \left(\frac{l}{2} + a\right)$, that is at point E

$$M_E = \frac{w}{8}(l^2 - 4a^2)$$

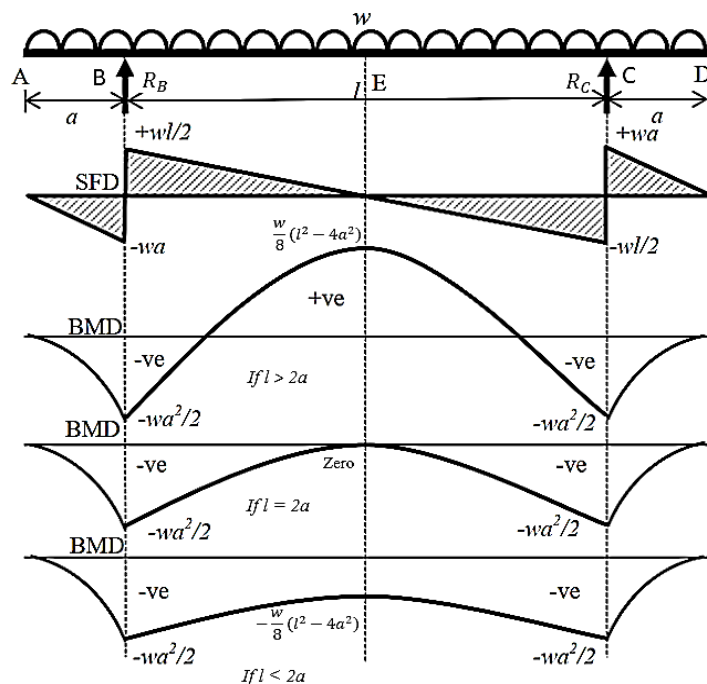
Note: Shear force diagram will remain same irrespective of the ratio of the l and a . However, from the above equation it can be seen that the bending moment will be affected by the ratio of the l and a

Case a: If $l > 2a$: $M_E = \frac{w}{8}(l^2 - 4a^2) = \text{Positive}$; i.e., $M_E > 0$

(1 mark)

Case b: If $l = 2a$: $M_E = \frac{w}{8}(4a^2 - 4a^2) = 0$; i.e., $M_E = 0$

Case c: If $l < 2a$: $M_E = \frac{w}{8}(l^2 - 4a^2) = \text{Negative}$; i.e., $M_E < 0$



(2 marks)

5. **Figure 5** shows the dual-grip clamp used to provide added clamping force with a positive action. If the vertical screw is tightened to produce a clamping force of 3 kN and then the horizontal screw is tightened until the force in the screw at A is doubled,
6. Draw the Free Body Diagram of the clamping system. [3 Marks]
7. Find the force acting on the horizontal screw [2 Marks]
8. Find the force acting on the vertical screw [2 Marks]
9. Find the total reaction R on the pin at B. [3 Marks]

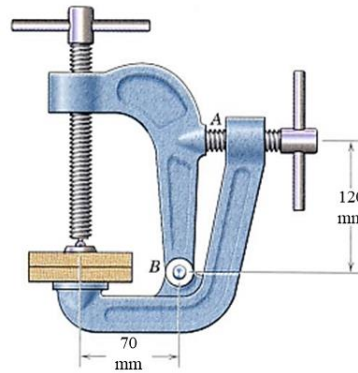


Figure 5

Solution of Q5:

Step 1: Drawing the Free Body Diagram as represented in Figure 5(a) of given Figure 5.

[3 Marks]

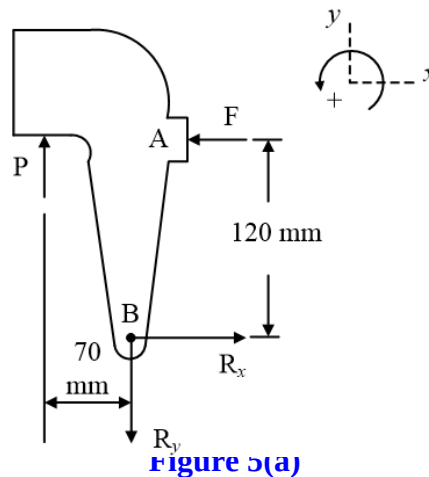


Figure 5(a)

Step 2: Force acting on the horizontal screw of the dual-grip clamping system [2 Marks]

For $P=3\text{ kN}$:

Taking summation of moments about point B equals to zero ($\sum M_B = 0$), we have:

$$\sum M_B = 0: \quad 120F - 3(70) = 0, \quad F = 1.75\text{ kN}$$

Step 3: Force acting on the vertical screw of the dual-grip clamping system [2 Marks]

$$\text{For } F = 2(1.75) = 3.5\text{ kN}, \quad P = 3(2) = 6\text{ kN}$$

Step 4: The total reaction R on the pin at point B

[3 Marks]

To get the reaction R_x by summing the forces along x -direction equals zero ($\sum F_x = 0$), we have:

$$\sum F_x = 0: \quad R_x - 3.5 = 0, \quad R_x = 3.5 \text{ kN}$$

To get the reaction R_y by summing the forces along y -direction equals zero ($\sum F_y = 0$), we have:

$$\sum F_y = 0: \quad -R_y + 6 = 0, \quad R_y = 6 \text{ kN}$$

So, the total reaction R at point B can be found out by:

$$R = \sqrt{(3.5)^2 + (6)^2} = 6.94 \text{ kN}$$